

17.To Implement how to List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State Information, Test Job Listings.

To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching.

```
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t jobs[3];

    printf("Starting background jobs...\n");

    for (int i = 0; i < 3; i++) {
        jobs[i] = fork();

        if (jobs[i] == 0) {
            printf("Job %d running, PID = %d\n",
                i + 1, getpid());

            sleep(5);

            exit(0);
        }
    }

    printf("\nActive Jobs:\n");

    for (int i = 0; i < 3; i++) {
        printf("[%d] PID = %d - Running\n",
            i + 1, jobs[i]);
    }

    for (int i = 0; i < 3; i++)
        waitpid(jobs[i], NULL, 0);

    printf("\nAll jobs completed.\n");

    return 0;
}
```

```
bindu@Bindusree:~$ nano skill17.c
bindu@Bindusree:~$ gcc skill17.c -o skill17
bindu@Bindusree:~$ ./skill17
Starting background jobs...
Job 1 running, PID = 773
Job 2 running, PID = 774

Active Jobs:
[PID = 1] PID = 773 - Running
Job 3 running, PID = 775
[PID = 2] PID = 774 - Running
[PID = 3] PID = 775 - Running

All jobs completed.
bindu@Bindusree:~$ |
```